ITRI626

ARTIFICIAL INTELLIGENCE

Faculty of Natural and Agricultural Science

Study guide compiled by: J.J. Greeff, M. Kisten and A.E. Ezugwu

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North-West University



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It all starts here

- Ranked in the top 5% of universities globally by the QS-rankings
- Contribute the second largest of number of graduates annually to the labour market

Dit begin alles hier

- As een van die top 5% universiteite wêreldwyd deur die QS-ranglys aangewys
- Lewer jaarliks die tweede-meeste graduandi aan die arbeidsmark

Gotlhe go simolola fano

- Re beilwe mo gare ga diyunibesiti tse 5% tse di kwa godimo go ya ka peo ya maemo ya QS
- Ngwaga le ngwaga go abelwa palo ya bobedi ka bogolo ya badiri mo maketeng ya badiri

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Module information

Module code	ITRI626		
Module Credits	16		
Prerequisite	None		
Module name	Artificial Intelligence II		
	Potchefstroom	Vaal Triangle	
Name of lecturer(s)	Prof A. E. Ezugwu	Mr. Petri Jooste	
Office telephone	+27182994359	+27169103264	
Email address	Absalom.Ezugwu@nwu.ac.za	Petri.Jooste@nwu.ac.za	
Building and Office nr	Building G3, room G16		
Consulting hours	Information on consulting hours will be communicated via eFundi.		

Lecturer information for distance students will be communicated electronically.

Welcome

In this module, we will explore deeper into the field of machine learning and logical agents—two key components that enable machines to exhibit intelligent behaviour. Building on the foundational knowledge you have acquired in previous AI modules, we will focus on how data can be used to train systems to perform tasks they were not explicitly programmed to do. Through this, you will better understand how learning algorithms and agent-based reasoning contribute to the development of intelligent systems.

Rationale

Artificial Intelligence is busy crossing the bridge from experimental technology in the laboratory to accepted technology in the commercial world. It is therefore of the utmost importance that you get acquainted with the technology so that you can make an informed decision when Artificial Intelligence methods are proposed as solutions to problems.

On the successful completion of this module, you will have a working knowledge of how to use a variety of machine learning algorithms as well as deep neural networks to solve data-driven problems. You will learn how to build a neural network from the ground up to

understand how frameworks are constructed in industry, as well as how to use some of the most common frameworks to solve problems using machine learning.

Outcomes

On completion of this module, students should be able to:

- Apply machine learning algorithms to build predictive models
- Understand and be able to apply regression models
- Understand and be able to apply classification models
- Explain the biological inspiration for neural networks
- Discuss and apply neural network models and architectures to solve problems
- Understand neural networks from three levels of abstraction
    - The operation of a neural network framework
    - The building of neural networks using frameworks
    - Transfer learning to reuse models from different domains
- Communicate effectively orally and in writing
- Be able to reason about the ethical implications of machine learning

Module Plan

The module plan will be made available on eFundi. It contains the schedule we'll follow to complete the study units as well as important dates to remember.

Assessment

Medical certificates and other valid excuses for tests and/or assignments must be submitted to the relevant lecturer/secretary within 7 days after the test was written or the assignment had to be submitted. Failure to present excuses in time will result in the excuse not being accepted and a nil mark will be awarded.

Enquiries about marks or requests for remarking tests/assignments must be directed to the relevant lecturer/secretary or submitted to him/her within 7 days after the test/assignment has been returned to students. Failure to act in time will result in incorrect calculations/entries not being corrected, or that tests/assignments will not be remarked.

To successfully complete the module, you will need to keep the following in mind:

- Participation mark: a minimum of 40% to obtain admission to the examination.
- You will always get two examination opportunities to complete your examination (However, this exception does not apply to modules that are assessed entirely through continuous assessment).
- Examination sub-minimum: 40%
- You require a final module mark of at least 50% to pass the module.
- 75% and higher indicates a distinction.

Assessment plan:

A) Assessments that count towards your participation mark (Formative)

Scheduled tests and theoretical and practical assignments will be given throughout the semester. These will contribute towards a participation mark. You will receive information Via eFundi on when and how to complete these tests and assignments. The emphasis will be on practical projects/assignments that are accompanied by written reports on the work you have done.

B) Examination (Summative assessment)
There will be no final examination. Instead, assessment will follow a continuous assessment model, consisting of three or more major scheduled tests and ongoing mini-projects throughout the semester.

C) Informal assessment that will support and give feedback on your learning (Formative assessment)
(any/all of these may also be included in calculating your participation mark)
Class assignments, class tests and quizzes may be given during sessions. Quizzes for self-assessment may also be available in the course.

Examination admission:

A module mark of at least 40% as well as satisfactory attendance of facilitation sessions must be attained to qualify for entrance to the exam. In the exam, a minimum mark of 40% must be attained, and a final mark of at least 50% is required to pass the course.

Calculation of participation marks for students will be announced during lectures.

Study materials

Russel, S. and Norvig, P. 2010 Artificial Intelligence A Modern Approach. Prentice Hall. Third Edition. ISBN: 9781292153964

Trask, A. 2019 Grokking Deep Learning, Manning, First Edition. ISBN: 9781617293702

Chollet, F 2021 Deep Learning with Python, Manning, Second Edition. ISBN: 9781617296864

Additional Sources

Luger, G.F. 2008. Artificial Intelligence, Structures and Strategies for Complex Problem Solving. Harlow, Addison-Wesley, Sixth Edition. ISBN: 9780321545893

Ertel, W. 2017. Introduction to Artificial Intelligence. London: Springer International Publishing. Second Edition. ISBN: 9783319584867

How to study

The following method is suggested:

1. **First, read through the material** in your study guide or the prescribed material and indicate the most important parts.
2. **Systematise** the work in accordance with the **learning outcomes** as set out in the **study guide** and summarise the material in the light thereof.

3. The **learning outcomes** help you to identify the most important sections of the work. The other information is not unimportant as it is relevant to the study outcomes. You only have to have a cursory view thereof for exam purposes. Do not, however, ignore this material. You should be able to place your knowledge in context.
4. **Thereafter complete all the learning activities indicated on eFundi.**
5. **Always read** through your own answers **again** before comparing them to the model answers given.
6. **Participate** in the group discussions and make use of the facilitator's expertise.
7. Make sure that you **understand** the definitions, methods and techniques;
8. **Work** through the examples to see whether you can apply the technique **practically**;
9. Complete **all** exercises and assignments communicated to you by the lecturer;
10. Complete **all** self-evaluation exercises and monitor your own progress.

It is important in this module that you can practically apply every method and technique that you acquire. The final test to see whether you understand a technique or method is therefore not whether you can write down the technique or method verbatim, but whether you can use it to solve a problem successfully.

The exercises serve as a test to see whether you understand the work in the study unit/study section and whether you can practically apply it to problems.

Icons



Time allocation



Learning outcomes



Study material



Assessment /
Assignments



Individual exercise



Group Activity



Example



Reflection

Warning against plagiarism

Plagiarism is a serious offence and you should familiarise yourself with the plagiarism policy of the NWU: http://library.nwu.ac.za/copyright-and-plagiarism.

Please refer to the Policy on Academic Integrity which is found on the following website: http://www.nwu.ac.za/content/policy_rules

Study unit 1
MACHINE LEARNING

Study hours



The time scheduled for this study unit is 4 hours.

Learning outcomes



At the end of this study unit, you should:

- Describe the concept of learning and learning approaches
- Describe the difference between parametric and non-parametric learning
- Describe the difference between machine learning paradigms in relation to how they treat data and their application spaces.
- Design and analyse machine learning algorithms

Study material



- Study pp 693 - 697 in Russel and Norvig and chapters 1 and 2 of Trask. Pay attention to the different paradigms of machine learning and the difference between parametric and non-parametric learning.

Exercises



- Your lecturer will communicate exercises and submission instructions.

Study unit 2

NEURAL NETWORKS - INTRODUCTION

Study hours



The time scheduled for this study unit is 16 hours.

Learning outcomes



At the end of this study unit, you should:

- Understand the basic mathematical model underlying the operation of a single neuron in a neural network
- Be able to describe
    - Single-input, single-output network
    - Multiple input, single output network
    - Single input, multiple output network
    - Multiple input, multiple output network

Study material



Study chapter 3 of Trask and build the model along with the practical examples given. Building the model in Python will give you a first-hand understanding of the network structure and the operations performed by the neurons in each of the neural network layers.

Exercises



- Your lecturer will communicate exercises and submission instructions.

Study unit 3
LEARNING MODELS – BASIC APPLICATIONS

Study hours



The time scheduled for this study unit is 20 hours.

Learning outcomes



At the end of this study unit, you should:

- Describe the black box model of machine learning algorithms
- Discuss the training, validation and testing functions of data
- Discuss the gradient descent algorithm for training models
- Discuss different neuron activation functions and their purposes
- Understand and be able to create regression models to solve problems using
    - A linear regression model using Scikit-Learn
    - Create a neural network to solve similar problems
- Understand and be able to create classification models to solve problems using
    - Support vector classifier model using Scikit-Learn
    - Create a neural network to solve similar problems
- Describe the full structure of a basic deep neural network

Study material



- Study chapters 4 and 5 in Trask and follow along with the practical demonstrations that will be performed in Python. You should be able to create simple classification and regression models using both neural networks and other machine learning algorithms on the standard benchmark datasets. Study pp 718 – 727 and 744 – 748 in Russel and Norvig to understand the functioning of Regression models and Support Vector Machines.

Exercises



- Your lecturer will communicate exercises and submission instructions.

Study unit 4

CONVOLUTIONAL NEURAL NETWORKS

Study hours



The time scheduled for this study unit is 20 hours.

Learning outcomes



At the end of this study unit, you should:

- Discuss the application areas of convolutional neural networks
- Discuss the function of a convolutional layer
- Discuss the function of a pooling layer

Study material



- Study the handouts, slides, and video on convolutional neural networks that your lecturer will distribute. You are encouraged to also read up on ImageNet and the different networks that have been used against this image classification benchmark. Summarise your findings for discussion in class.

Exercises



- Your lecturer will communicate exercises and submission instructions.

Study unit 5
DIFFERENT FORMS OF LEARNING

Study hours



The time scheduled for this study unit is 20 hours.

Learning outcomes



At the end of this study unit, you should:

- Be able to recognise the difference between using labelled and unlabelled data
- Understand the nature of data and how we can apply different techniques to unlabelled data than what we can to labelled data.
- Be able to perform clustering operations on unlabelled data
- Understand and be able to use tools that use semi-supervised learning

Study material



- Study the handouts, slides, and video on convolutional neural networks that your lecturer will distribute. Summarise your findings for discussion in class.

Exercises



- Your lecturer will communicate exercises and submission instructions.

Study unit 6
TRANSFER LEARNING

Study hours



The time scheduled for this study unit is 20 hours.

Learning outcomes



At the end of this study unit, you should:

- Understand and be able to use transfer learning to apply a model that has been trained in one domain to a different domain.

Study material



- Study the handouts, slides, and video on convolutional neural networks that your lecturer will distribute. This section of the course will be handled predominantly practically. Still, you are encouraged to seek as many sources as possible to contribute to your understanding of transfer learning beyond what will be presented in class and in the practical sessions.

Exercises



- Your lecturer will communicate exercises and submission instructions.

Study unit 7
LARGE LANGUAGE MODELS

Study hours



The time scheduled for this study unit is 20 hours.

Learning outcomes



At the end of this study unit, you should:

- Understand the basic building blocks and operation of Large Language Models
- Be able to use Large Language Models effectively as they exist already
- Be able to fine-tune a Large Language Model to a specific domain

Study material



- Study the handouts, slides, and video on large language models that your lecturer will distribute. This section of the course will be handled predominantly practically. Still, you are encouraged to seek as many sources as possible to contribute to your understanding beyond what will be presented in class and in the practical sessions.

Exercises



- Your lecturer will communicate exercises and submission instructions.